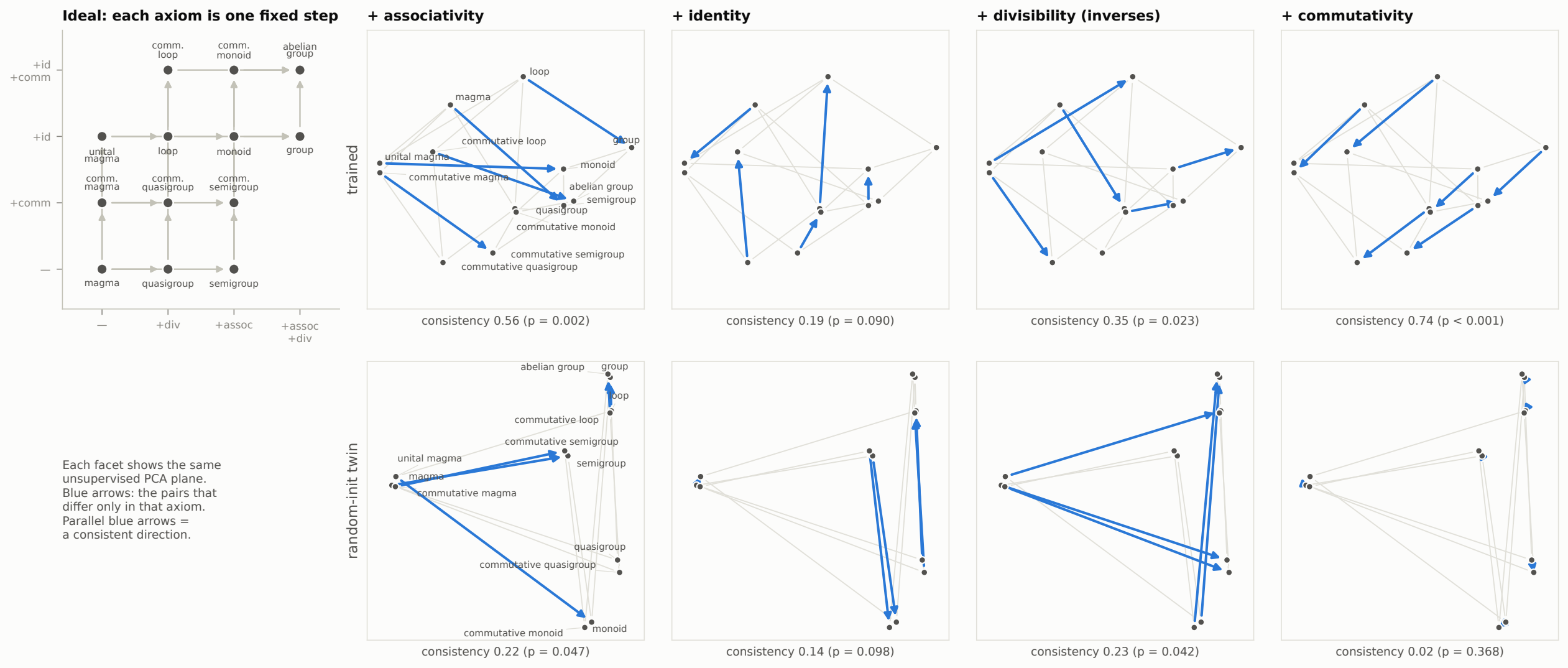
Ideal: each axiom is one fixed step
+ associativity
+ identity
+ divisibility (inverses)
+ commutativity
+id +comm
+id
+comm
—
comm. loop
comm. monoid
abelian group
unital magma
loop
monoid
group
comm. magma
comm. quasigroup
comm. semigroup
magma
quasigroup
semigroup
—
+div
+assoc
+assoc +div
trained
random-init twin
consistency 0.56 (p = 0.002)
consistency 0.19 (p = 0.090)
consistency 0.35 (p = 0.023)
consistency 0.74 (p < 0.001)
consistency 0.22 (p = 0.047)
consistency 0.14 (p = 0.098)
consistency 0.23 (p = 0.042)
consistency 0.02 (p = 0.368)
Each facet shows the same unsupervised PCA plane. Blue arrows: the pairs that differ only in that axiom. Parallel blue arrows = a consistent direction.